Amplifier Frequency Response

Electronic Devices and Circuits Study Report

Based on 24EECE2001 Lecture Material

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1 Transfer Functions and Pole-Zero Concepts

A transfer function describes how a linear system modifies an input signal to produce an output signal in the complex frequency domain. It is generally denoted by $H(s)$, where s is the complex frequency variable ($s = \sigma + j\omega$).

1.1 Poles and Zeros

A transfer function can typically be expressed as the ratio of two polynomials:

$$H(s) = \frac{\text{Numerator}(s)}{\text{Denominator}(s)} \tag{1}$$

- **Zeros:** The roots of the numerator polynomial. These are the values of s that make the transfer function equal to zero. In a Bode magnitude plot, a zero corresponds to a corner frequency at which the slope of the response increases by $+20$ dB/decade.
- **Poles:** The roots of the denominator polynomial. These are the values of s that make the transfer function approach infinity. Poles determine the stability and transient behavior of the system. In a Bode magnitude plot, a pole corresponds to a corner frequency where the slope decreases by -20 dB/decade.

2 First-Order RC Circuits and Frequency Response

First-order RC (Resistor-Capacitor) circuits form the foundation for understanding the frequency-dependent behavior of complex amplifier models.

2.1 RC Low-Pass Filter

A low-pass filter (LPF) allows low-frequency signals to pass with minimal attenuation while attenuating high-frequency signals.

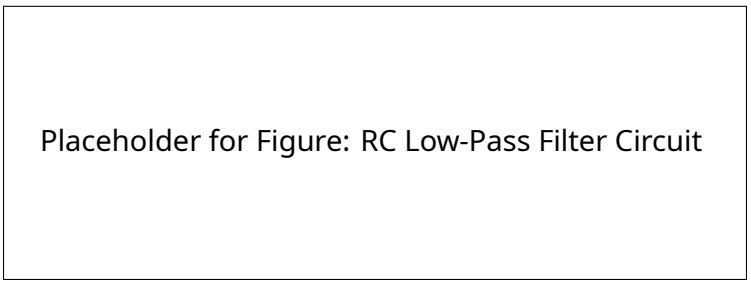


Figure 1: First-order RC Low-Pass Filter

The transfer function $H(s) = \frac{V_o(s)}{V_i(s)}$ is derived using the voltage divider rule:

$$H(s) = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{1}{1 + sRC} \tag{2}$$

The numerator is a constant (1), meaning there are no finite zeros. The denominator has a pole at $s = -1/RC$. For sinusoidal steady-state analysis, we substitute $s = j\omega$:

$$H(j\omega) = \frac{1}{1 + j\omega RC} \tag{3}$$

The magnitude response is given by:

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}} \tag{4}$$

The cutoff frequency, ω_c, occurs when the magnitude drops to 0.707 (or -3 dB) of its DC value. This happens when $\omega_c RC = 1$, leading to:

$$\omega_c = \frac{1}{RC} \quad \Longrightarrow \quad f_c = \frac{1}{2\pi RC} \tag{5}$$

2.2 RC High-Pass Filter

A high-pass filter (HPF) attenuates low frequencies and allows high frequencies to pass.

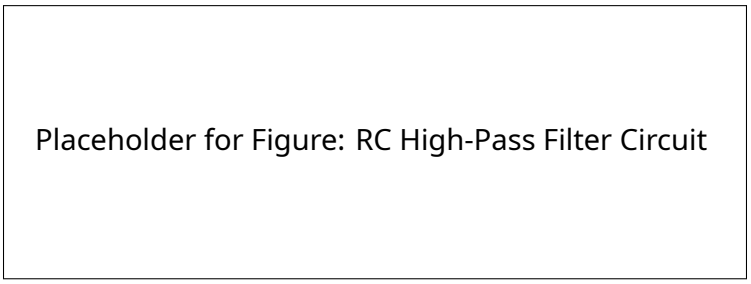


Figure 2: First-order RC High-Pass Filter

The transfer function is derived as:

$$H(s) = \frac{R}{R + \frac{1}{sC}} = \frac{sRC}{1 + sRC} \tag{6}$$

Setting the numerator to zero gives a zero at $s = 0$. The denominator gives a pole at $s = -1/RC$. Substituting $s = j\omega$:

$$H(j\omega) = \frac{j\omega RC}{1 + j\omega RC} \tag{7}$$

The magnitude response is:

$$|H(j\omega)| = \frac{\omega RC}{\sqrt{1 + (\omega RC)^2}} \tag{8}$$

Similar to the low-pass filter, the cutoff frequency is:

$$f_c = \frac{1}{2\pi RC} \tag{9}$$

3 Capacitive Effects in PN Junctions

A PN junction exhibits capacitive effects that significantly influence the high-frequency performance of semiconductor devices.

3.1 Depletion or Junction Capacitance

When a PN junction is reverse-biased, a depletion region forms. The width of this region depends on the doping concentrations, N_A (acceptor) and N_D (donor). The depletion region extends deeper into the more lightly doped material to uncover an equal amount of charge on both sides.

Let x_p and x_n be the widths of the depletion region in the p-side and n-side, respectively. The charge equality condition is:

$$qAx_nN_D = qAx_pN_A \quad \Longrightarrow \quad \frac{x_n}{x_p} = \frac{N_A}{N_D} \tag{10}$$

The total width $W = x_n + x_p$ is given by:

$$W = \sqrt{\frac{2\epsilon_s}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)V_0} \tag{11}$$

where ϵ_s is the permittivity of silicon ($\approx 1.04 \times 10^{-12}$ F/cm) and V_0 is the built-in voltage.

The stored charge Q_J on either side is:

$$Q_J = Aq\left(\frac{N_AN_D}{N_A+N_D}\right)W = A\sqrt{2\epsilon_s q\left(\frac{N_AN_D}{N_A+N_D}\right)(V_0+V_R)} = \alpha\sqrt{V_0+V_R} \tag{12}$$

The **junction capacitance**, C_j, relates the change in charge to the change in reverse voltage V_R:

$$C_j = \frac{dQ_J}{dV_R} = \frac{\alpha}{2\sqrt{V_0+V_R}} \tag{13}$$

For a general graded junction, this can be written as:

$$C_j = \frac{C_{j0}}{\left(1+\frac{V_R}{V_0}\right)^m} \tag{14}$$

where C_{j0} is the zero-bias junction capacitance and m is the grading coefficient ($1/3$ to $1/2$).

3.2 Diffusion Capacitance

Diffusion capacitance occurs when the PN junction is forward-biased, causing minority carriers to be injected across the junction. The excess minority carrier charge stored in the n-region (Q_p) is related to the forward current I_p and the hole lifetime τ_p:

$$Q_p = I_p\tau_p \tag{15}$$

The total stored charge is $Q = \tau_T I$, where τ_T is the overall mean transit time. The **diffusion capacitance** is defined as the rate of change of this charge with respect to the forward voltage V:

$$C_d = \frac{dQ}{dV} = \frac{\tau_T}{V_T}I \tag{16}$$

Because C_d is proportional to the forward current I, it is significant in forward bias but negligibly small in reverse bias.

4 Internal Capacitive Effects and the High-Frequency MOSFET Model

4.1 MOSFET Capacitances

The high-frequency behavior of a MOSFET is limited by its internal capacitances, which fall into two categories:

1. **Gate Capacitive Effect:** The polysilicon gate, oxide dielectric, and channel form a parallel-plate capacitor.
2. **Junction Capacitances:** The reverse-biased PN junctions between the source/drain diffusions and the substrate body form depletion-layer capacitances (C_{sb} and C_{db}).

The gate capacitance is distributed between the source and drain depending on the region of operation:

- **Triode Region:** The channel depth is relatively uniform.

$$C_{gs} = C_{gd} = \frac{1}{2} WLC_{ox} \tag{17}$$

- **Saturation Region:** The channel is pinched off at the drain end.

$$C_{gs} = \frac{2}{3} WLC_{ox}, \quad C_{gd} = 0 \text{ (idealized)} \tag{18}$$

  (Note: In practice, overlap capacitance ensures C_{gd} is never truly zero and plays a critical role in high-frequency response).

- **Cutoff Region:** The channel disappears entirely.

$$C_{gs} = C_{gd} = 0, \quad C_{gb} = WLC_{ox} \tag{19}$$

4.2 The MOSFET Unity-Gain Frequency (f_T)

A key figure of merit for high-frequency amplifiers is the unity-gain frequency f_T, defined as the frequency at which the short-circuit current gain of the common-source configuration drops to unity.

Applying a current signal I_i to the gate with the drain short-circuited to AC ground:

$$I_i = s(C_{gs} + C_{gd})V_{gs} \tag{20}$$

The short-circuit output current I_o is approximately $g_m V_{gs}$ (neglecting the small feed-forward current through C_{gd}). The current gain is:

$$\frac{I_o}{I_i} \approx \frac{g_m}{s(C_{gs} + C_{gd})} \tag{21}$$

Substituting $s = j\omega$ and setting the magnitude to 1 at $\omega = \omega_T$:

$$\frac{g_m}{\omega_T(C_{gs} + C_{gd})} = 1 \quad \Longrightarrow \quad f_T = \frac{g_m}{2\pi(C_{gs} + C_{gd})} \tag{22}$$

A higher f_T indicates a transistor capable of operating at higher frequencies.

5 High-Frequency Response of Common-Source Amplifiers

5.1 Miller's Theorem

Miller's theorem states that an impedance Z connected between two nodes (Node 1 and Node 2), where the voltage gain from Node 1 to Node 2 is $K = V_2/V_1$, can be replaced by two equivalent impedances to ground:

$$Z_1 = \frac{Z}{1-K}, \quad Z_2 = \frac{Z}{1-1/K} \tag{23}$$

5.2 The Common-Source (CS) Amplifier at High Frequencies

In a common-source amplifier, the gate-to-drain capacitance C_{gd} bridges the input (gate) and output (drain). The voltage gain from gate to drain is large and negative, given by the midband gain:

$$A_M = -g_m R'_L \tag{24}$$

where $R'_L = r_o||R_D||R_L$.

Using Miller's theorem, the bridging capacitance C_{gd} is reflected to the input side as an equivalent capacitance C_{eq}:

$$C_{eq} = C_{gd}(1 - A_M) = C_{gd}(1 + g_m R'_L) \tag{25}$$

This phenomenon, where a small physical capacitance C_{gd} is multiplied by the amplifier's gain to create a massive input capacitance, is known as the **Miller Effect**.

The total equivalent input capacitance is:

$$C_{in} = C_{gs} + C_{eq} = C_{gs} + C_{gd}(1 + g_m R'_L) \tag{26}$$

The high-frequency response of the CS amplifier is dominated by the low-pass filter formed by the source resistance and this massive input capacitance. The upper 3-dB cutoff frequency, f_H, is determined by the time constant $C_{in}R'_{sig}$ (where $R'_{sig} = R_{sig}||R_G \approx R_{sig}$):

$$f_H = \frac{1}{2\pi C_{in} R'_{sig}} \tag{27}$$

Because the Miller effect inflates C_{in}, the CS amplifier typically suffers from a relatively low upper-cutoff frequency (f_H), limiting its bandwidth.

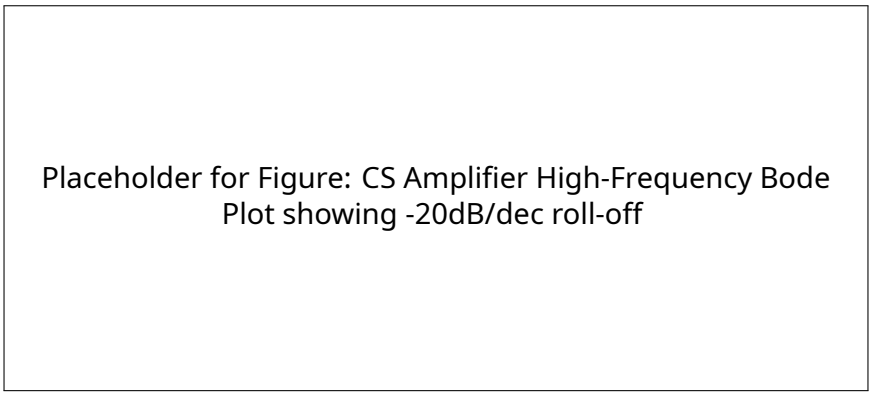


Figure 3: Frequency response magnitude plot for the CS amplifier

6 Consolidated Formula Sheet

Concept	**Equation**
RC Filter Cutoff Frequency	$f_c = \frac{1}{2\pi RC}$
Junction Capacitance	$C_j = \frac{C_{j0}}{\left(1+\frac{V_R}{V_0}\right)^m}$
Diffusion Capacitance	$C_d = \frac{\tau_T}{V_T} I$
MOSFET Unity-Gain Freq.	$f_T = \frac{g_m}{2\pi(C_{gs}+C_{gd})}$
Miller Equivalent Input Cap.	$C_{eq} = C_{gd}(1 + g_m R'_L)$
CS Total Input Capacitance	$C_{in} = C_{gs} + C_{gd}(1 + g_m R'_L)$
CS Upper Cutoff Freq.	$f_H = \frac{1}{2\pi C_{in} R'_{sig}}$

Table 1: Key Equations for Amplifier Frequency Response